Octal D Flip-Flop with 3-State Outputs

MC74AC574, MC74ACT574

The MC74AC574/74ACT574 is a high-speed, low power octal flip-flop with a buffered common Clock (CP) and a buffered common Output Enable ($\overline{OE}$). The information presented to the D inputs is stored in the flip-flops on the LOW-to-HIGH Clock (CP) transition.

The MC74AC574/74ACT574 is functionally identical to the MC74AC374/74ACT374 except for the pinouts.

Features

- Inputs and Outputs on Opposite Sides of Package Allowing Easy Interface with Microprocessors
- Useful as Input or Output Port for Microprocessors
- Functionally Identical to MC74AC374/74ACT374
- 3-State Outputs for Bus-Oriented Applications
- Outputs Source/Sink 24 mA
- 'ACT574 Has TTL Compatible Inputs
- Pb-Free Packages are Available

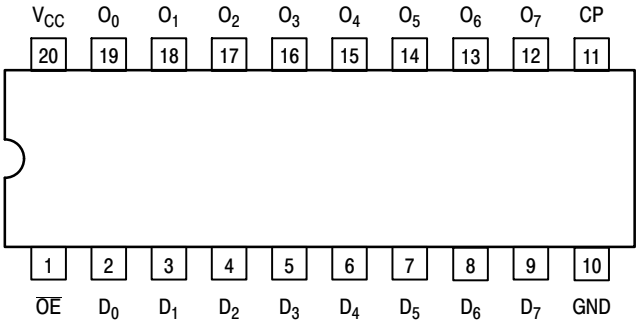


Figure 1. Pinout: 20-Lead Packages Conductors
(Top View)

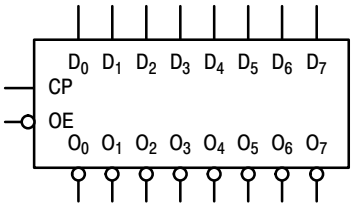
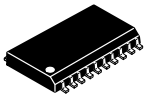


Figure 2. Logic Symbol

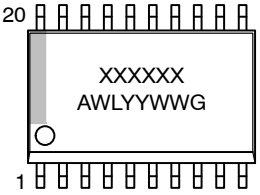
PIN ASSIGNMENT

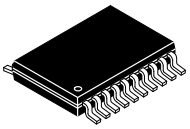
Pin	Function
D_0–D_7	Data Inputs
CP	Clock Pulse Input
$\overline{OE}$	3-State Output Enable Input
O_0–O_7	3-State Outputs

MARKING DIAGRAMS

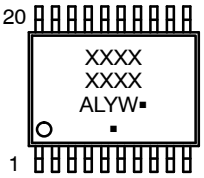


SOIC-20W
DW SUFFIX
CASE 751D





TSSOP-20
DT SUFFIX
CASE 948E



XXXXXX = Specific Device Code
A = Assembly Location
WL, L = Wafer Lot
YY, Y = Year
WW, W = Work Week
G or ▪ = Pb-Free Package
(Note: Microdot may be in either location)

ORDERING INFORMATION

See detailed ordering and shipping information on page 7 of this data sheet.

© Semiconductor Components Industries, LLC, 2006

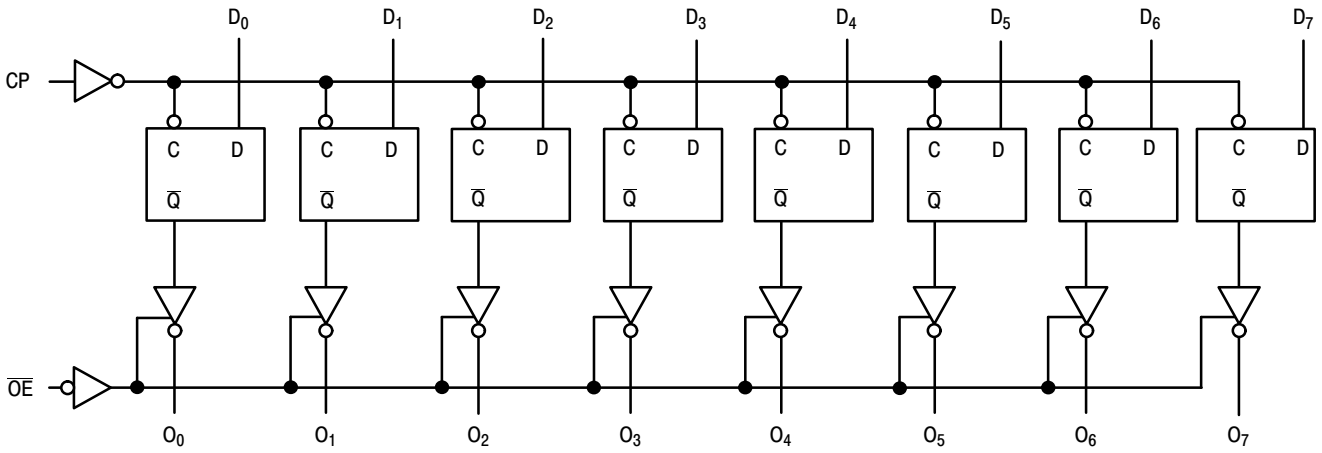
FUNCTIONAL DESCRIPTION

The MC74AC574/74ACT574 consists of eight edge-triggered flip-flops with individual D-type inputs and 3-state true outputs. The buffered clock and buffered Output Enable are common to all flip-flops. The eight flip-flops will store the state of their individual D inputs that meet the setup and hold time requirements on the LOW-to-HIGH Clock (CP) transition. With the Output Enable ($\overline{OE}$) LOW, the contents of the eight flip-flops are available at the outputs. When $\overline{OE}$ is HIGH, the outputs go to the high impedance state. Operation of the $\overline{OE}$ input does not affect the state of the flip-flops.

FUNCTION TABLE

Inputs			Internal	Outputs	
OE	CP	D	Q	O_n	Function
H	H	L	NC	Z	Hold
H	H	H	NC	Z	Hold
H	⌐	L	L	Z	Load
H	⌐	H	H	Z	Load
L	⌐	L	L	L	Data Available
L	⌐	H	H	H	Data Available
L	H	L	NC	NC	No Change in Data
L	H	H	NC	NC	No Change in Data

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial
Z = High Impedance
⌐ = LOW-to-HIGH Clock Transition
NC = No Change



NOTE: This diagram is provided only for the understanding of logic operations and should not be used to estimate propagation delays.

Figure 3. Logic Diagram

Share Feedback
Your Opinion Matters

MAXIMUM RATINGS

Symbol	Parameter	Value	Unit
V_{CC}	DC Supply Voltage (Referenced to GND)	−0.5 to +6.5	V
V_{IN}	DC Input Voltage (Referenced to GND)	−0.5 to V_{CC} +0.5	V
V_{OUT}	DC Output Voltage (Referenced to GND)	−0.5 to V_{CC} +0.5	V
I_{IN}	DC Input Current, per Pin	±20	mA
I_{OUT}	DC Output Sink/Source Current, per Pin	±50	mA
I_{CC}	DC V_{CC} or GND Current per Output Pin	±50	mA
T_{stg}	Storage Temperature	−65 to +150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter		Min	Typ	Max	Unit
V_{CC}	Supply Voltage	'AC	2.0	5.0	6.0	V
		'ACT	4.5	5.0	5.5	
V_{IN}, V_{OUT}	DC Input Voltage, Output Voltage (Ref. to GND)		0	–	V_{CC}	V
t_r, t_f	Input Rise and Fall Time (Note 1) 'AC Devices except Schmitt Inputs	V_{CC} @ 3.0 V	–	150	–	ns/V
		V_{CC} @ 4.5 V	–	40	–	
		V_{CC} @ 5.5 V	–	25	–	
t_r, t_f	Input Rise and Fall Time (Note 2) 'ACT Devices except Schmitt Inputs	V_{CC} @ 4.5 V	–	10	–	ns/V
		V_{CC} @ 5.5 V	–	8.0	–	
T_A	Operating Ambient Temperature Range		−40	25	85	°C
I_{OH}	Output Current − High		–	–	−24	mA
I_{OL}	Output Current − Low		–	–	24	mA

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

1. V_{IN} from 30% to 70% V_{CC}; see individual Data Sheets for devices that differ from the typical input rise and fall times.
2. V_{IN} from 0.8 V to 2.0 V; see individual Data Sheets for devices that differ from the typical input rise and fall times.

Share Feedback
Your Opinion Matters

DC CHARACTERISTICS

Symbol	Parameter	V_{CC} (V)	74AC T_A = +25 °C Typ	74AC T_A = −40 °C to +85 °C Guaranteed Limits		Unit	Conditions
V_{IH}	Minimum High Level Input Voltage	3.0	1.5	2.1	2.1	V	V_{OUT} = 0.1 V or V_{CC} − 0.1 V
		4.5	2.25	3.15	3.15		
		5.5	2.75	3.85	3.85		
V_{IL}	Maximum Low Level Input Voltage	3.0	1.5	0.9	0.9	V	V_{OUT} = 0.1 V or V_{CC} − 0.1 V
		4.5	2.25	1.35	1.35		
		5.5	2.75	1.65	1.65		
V_{OH}	Minimum High Level Output Voltage	3.0	2.99	2.9	2.9	V	I_{OUT} = −50 μA
		4.5	4.49	4.4	4.4		
		5.5	5.49	5.4	5.4		
						V	*V_{IN} = V_{IL} or V_{IH}
		3.0	−	2.56	2.46		−12 mA
		4.5	−	3.86	3.76		I_{OH} −24 mA
		5.5	−	4.86	4.76		−24 mA
V_{OL}	Maximum Low Level Output Voltage	3.0	0.002	0.1	0.1	V	I_{OUT} = 50 μA
		4.5	0.001	0.1	0.1		
		5.5	0.001	0.1	0.1		
						V	*V_{IN} = V_{IL} or V_{IH}
		3.0	−	0.36	0.44		12 mA
		4.5	−	0.36	0.44		I_{OL} 24 mA
		5.5	−	0.36	0.44		24 mA
I_{IN}	Maximum Input Leakage Current	5.5	−	±0.1	±1.0	μA	V_I = V_{CC}, GND
I_{OLD}	†Minimum Dynamic Output Current	5.5	−	−	75	mA	V_{OLD} = 1.65 V Max
I_{OHD}		5.5	−	−	−75	mA	V_{OHD} = 3.85 V Min
I_{CC}	Maximum Quiescent Supply Current	5.5	−	8.0	80	μA	V_{IN} = V_{CC} or GND

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

NOTE: Note: I_{IN} and I_{CC} @ 3.0 V are guaranteed to be less than or equal to the respective limit @ 5.5 V V_{CC}.

* All outputs loaded; thresholds on input associated with output under test.

† Maximum test duration 2.0 ms, one output loaded at a time.

Share Feedback
Your Opinion Matters

AC CHARACTERISTICS (For Figures and Waveforms – See AND8277/D at www.onsemi.com)

Symbol	Parameter	V_{CC}* (V)	74AC T_A = +25 °C, C_L = 50 pF			74AC T_A = –40 °C to +85 °C, C_L = 50 pF		Unit	Fig. No.
			Min	Typ	Max	Min	Max		
f_{max}	Maximum Clock Frequency	3.3	75	–	–	60	–	MHz	3–3
		5.0	95	–	–	85	–		
t_{PLH}	Propagation Delay CP to O_n	3.3	3.5	–	13.5	3.5	15	ns	3–6
		5.0	2.0	–	9.5	2.0	11		
t_{PHL}	Propagation Delay CP to O_n	3.3	3.5	–	12	3.5	13.5	ns	3–6
		5.0	2.0	–	8.5	2.0	9.5		
t_{PZH}	Output Enable Time	3.3	2.5	–	11	2.5	12	ns	3–7
		5.0	2.0	–	8.5	2.0	9.0		
t_{PZL}	Output Enable Time	3.3	3.0	–	10.5	3.5	11.5	ns	3–8
		5.0	1.5	–	8.0	2.0	9.0		
t_{PHZ}	Output Disable Time	3.3	4.0	–	12	4.0	13	ns	3–7
		5.0	2.0	–	9.5	2.0	10.5		
t_{PLZ}	Output Disable Time	3.3	2.0	–	9.0	2.5	10	ns	3–8
		5.0	1.5	–	7.5	1.5	8.5		

* Voltage Range 3.3 V is 3.3 V ±0.3 V.
Voltage Range 5.0 V is 5.0 V ±0.5 V.

AC OPERATING REQUIREMENTS

Symbol	Parameter	V_{CC}* (V)	74AC T_A = +25 °C, C_L = 50 pF	74AC T_A = –40 °C to +85 °C, C_L = 50 pF	Unit	Fig. No.
			Typ	Guaranteed Minimum		
t_s	Setup Time, HIGH or LOW D_n to CP	3.3	–	2.5	3.0	ns	3–9
		5.0	–	1.5	2.0		
t_h	Hold Time, HIGH or LOW D_n to CP	3.3	–	1.5	1.5	ns	3–9
		5.0	–	1.5	1.5		
t_w	CP Pulse Width HIGH or LOW	3.3	–	6.0	7.0	ns	3–6
		5.0	–	4.0	5.0		

* Voltage Range 3.3 V is 3.3 V ±0.3 V.
Voltage Range 5.0 V is 5.0 V ±0.5 V.

Share Feedback
Your Opinion Matters

DC CHARACTERISTICS

Symbol	Parameter	V_{CC} (V)	74ACT T_A = +25 °C Typ	74ACT T_A = –40 °C to +85 °C Guaranteed Limits		Unit	Conditions
V_{IH}	Minimum High Level Input Voltage	4.5	1.5	2.0	2.0	V	V_{OUT} = 0.1 V
		5.5	1.5	2.0	2.0		or V_{CC} – 0.1 V
V_{IL}	Maximum Low Level Input Voltage	4.5	1.5	0.8	0.8	V	V_{OUT} = 0.1 V
		5.5	1.5	0.8	0.8		or V_{CC} – 0.1 V
V_{OH}	Minimum High Level Output Voltage	4.5	4.49	4.4	4.4	V	I_{OUT} = –50 μA
		5.5	5.49	5.4	5.4		
						V	*V_{IN} = V_{IL} or V_{IH}
		4.5	–	3.86	3.76		I_{OH} –24 mA
		5.5	–	4.86	4.76		–24 mA
V_{OL}	Maximum Low Level Output Voltage	4.5	0.001	0.1	0.1	V	I_{OUT} = 50 μA
		5.5	0.001	0.1	0.1		
							*V_{IN} = V_{IL} or V_{IH}
		4.5	–	0.36	0.44	V	I_{OL} 24 mA
		5.5	–	0.36	0.44		24 mA
I_{IN}	Maximum Input Leakage Current	5.5	–	±0.1	±1.0	μA	V_I = V_{CC}, GND
ΔI_{CCT}	Additional Max. I_{CC}/Input	5.5	0.6		1.5	mA	V_I = V_{CC} – 2.1 V
I_{OZ}	Maximum 3-State Current	5.5	–	±0.5	±5.0	μA	V_I (OE) = V_{IL}, V_{IH} V_I = V_{CC}, GND V_O = V_{CC}, GND
I_{OLD}	†Minimum Dynamic Output Current	5.5	–	–	75	mA	V_{OLD} = 1.65 V Max
I_{OHD}		5.5	–	–	–75	mA	V_{OHD} = 3.85 V Min
I_{CC}	Maximum Quiescent Supply Current	5.5	–	8.0	80	μA	V_{IN} = V_{CC} or GND

* All outputs loaded; thresholds on input associated with output under test.
† Maximum test duration 2.0 ms, one output loaded at a time.

AC CHARACTERISTICS (For Figures and Waveforms – See AND8277/D at www.onsemi.com)

Symbol	Parameter	V_{CC}* (V)	74ACT T_A = +25 °C, C_L = 50 pF Min	Typ	Max	74ACT T_A = –40 °C to +85 °C, C_L = 50 pF Min	Max	Unit	Fig. No.
f_{max}	Maximum Clock Frequency	5.0	100	–	–	85	–	ns	3–3
t_{PLH}	Propagation Delay CP to O_n	5.0	2.5	–	11	2.0	12	ns	3–6
t_{PHL}	Propagation Delay CP to O_n	5.0	2.0	–	10	1.5	11	ns	3–6
t_{PZH}	Output Enable Time	5.0	2.0	–	9.5	1.5	10	ns	3–7
t_{PZL}	Output Enable Time	5.0	2.0	–	9.0	1.5	10	ns	3–8
t_{PHZ}	Output Disable Time	5.0	2.0	–	10.5	1.5	11.5	ns	3–7
t_{PLZ}	Output Disable Time	5.0	2.0	–	8.5	1.5	9.0	ns	3–8

* Voltage Range 5.0 V is 5.0 V ±0.5 V.

Share Feedback
Your Opinion Matters

AC OPERATING REQUIREMENTS

Symbol	Parameter	V_{CC}* (V)	74ACT T_A = +25 °C C_L = 50 pF		74ACT T_A = −40 °C to +85 °C C_L = 50 pF	Unit	Fig. No.
			Typ	Guaranteed Minimum			
t_s	Setup Time, HIGH or LOW D_n to CP	5.0	–	2.5	2.5	ns	3–9
t_h	Hold Time, HIGH or LOW D_n to CP	5.0	–	1.0	1.0	ns	3–9
t_w	CP Pulse Width HIGH or LOW	5.0	–	3.0	4.0	ns	3–6

* Voltage Range 3.3 V is 3.3 V ±0.3 V.
Voltage Range 5.0 V is 5.0 V ±0.5 V.

CAPACITANCE

Symbol	Parameter	Value Typ	Unit	Test Conditions
C_{IN}	Input Capacitance	4.5	pF	V_{CC} = 5.0 V
C_{PD}	Power Dissipation Capacitance	40	pF	V_{CC} = 5.0 V

ORDERING INFORMATION

Device	Marking	Package	Shipping†
MC74AC574DWG	AC574	SOIC-20	38 Units / Rail
MC74AC574DWR2G	AC574	SOIC-20	1000 / Tape & Reel
MC74ACT574DWG	ACT574	SOIC-20	38 Units / Rail
MC74ACT574DWR2G	ACT574	SOIC-20	1000 / Tape & Reel
MC74AC574DTR2G	AC 574	TSSOP-20	2500 / Tape & Reel
MC74ACT574DTR2G	ACT 574	TSSOP-20	2500 / Tape & Reel

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

Share Feedback
Your Opinion Matters

REVISION HISTORY

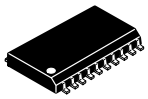
Revision	Description of Changes	Date
10	Modified Output Disable Time for 74AC across temperature range from 4.5 ns to 4.0 ns.	07/02/2025

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

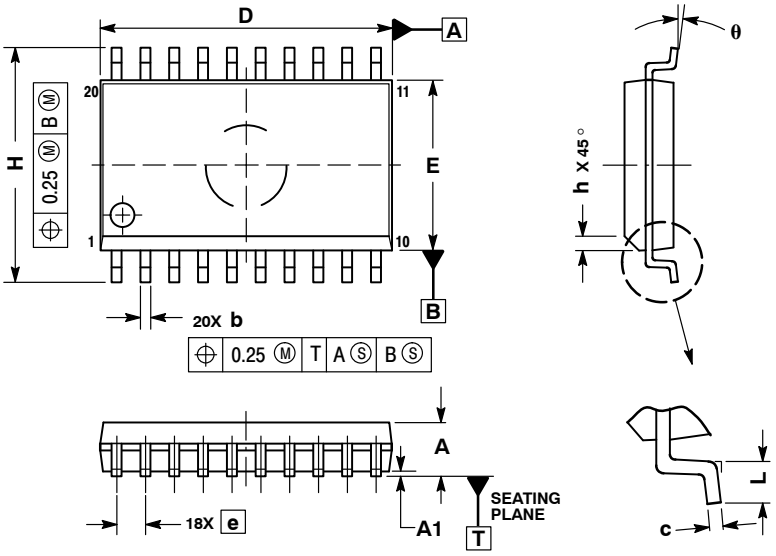
Share Feedback
Your Opinion Matters

SOIC−20 WB
CASE 751D−05
ISSUE H

DATE 22 APR 2015



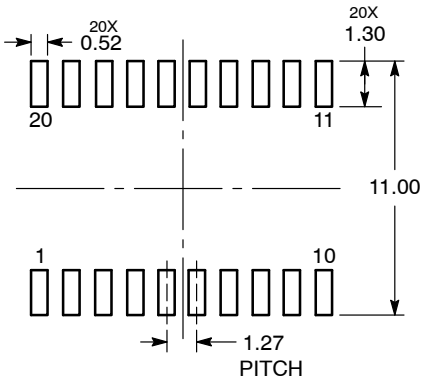
SCALE 1:1



NOTES:
1. DIMENSIONS ARE IN MILLIMETERS.
2. INTERPRET DIMENSIONS AND TOLERANCES PER ASME Y14.5M, 1994.
3. DIMENSIONS D AND E DO NOT INCLUDE MOLD PROTRUSION.
4. MAXIMUM MOLD PROTRUSION 0.15 PER SIDE.
5. DIMENSION B DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE PROTRUSION SHALL BE 0.13 TOTAL IN EXCESS OF B DIMENSION AT MAXIMUM MATERIAL CONDITION.

	MILLIMETERS	
DIM	MIN	MAX
A	2.35	2.65
A1	0.10	0.25
b	0.35	0.49
c	0.23	0.32
D	12.65	12.95
E	7.40	7.60
e	1.27 BSC	
H	10.05	10.55
h	0.25	0.75
L	0.50	0.90
θ	0°	7°

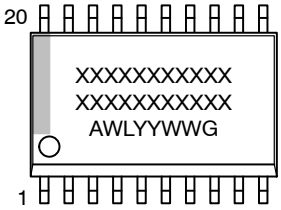
RECOMMENDED SOLDERING FOOTPRINT*



DIMENSIONS: MILLIMETERS

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



XXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week
G = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

DOCUMENT NUMBER:	**98ASB42343B**	Electronic versions are uncontrolled except when accessed directly from the Document Repository. Printed versions are uncontrolled except when stamped “CONTROLLED COPY” in red.	
DESCRIPTION:	**SOIC−20 WB**		**PAGE 1 OF 1**

onsemi and onsemi are trademarks of Semiconductor Components Industries, LLC dba **onsemi** or its subsidiaries in the United States and/or other countries. **onsemi** reserves the right to make changes without further notice to any products herein. **onsemi** makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does **onsemi** assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. **onsemi** does not convey any license under its patent rights nor the rights of others.

onsemi, ONSEMI , and other names, marks, and brands are registered and/or common law trademarks of Semiconductor Components Industries, LLC dba "**onsemi**" or its affiliates and/or subsidiaries in the United States and/or other countries. **onsemi** owns the rights to a number of patents, trademarks, copyrights, trade secrets, and other intellectual property. A listing of **onsemi**'s product/patent coverage may be accessed at www.onsemi.com/site/pdf/Patent−Marking.pdf. **onsemi** reserves the right to make changes at any time to any products or information herein, without notice. The information herein is provided "as−is" and **onsemi** makes no warranty, representation or guarantee regarding the accuracy of the information, product features, availability, functionality, or suitability of its products for any particular purpose, nor does **onsemi** assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. Buyer is responsible for its products and applications using **onsemi** products, including compliance with all laws, regulations and safety requirements or standards, regardless of any support or applications information provided by **onsemi**. "Typical" parameters which may be provided in **onsemi** data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. **onsemi** does not convey any license under any of its intellectual property rights nor the rights of others. **onsemi** products are not designed, intended, or authorized for use as a critical component in life support systems or any FDA Class 3 medical devices or medical devices with a same or similar classification in a foreign jurisdiction or any devices intended for implantation in the human body. Should Buyer purchase or use **onsemi** products for any such unintended or unauthorized application, Buyer shall indemnify and hold **onsemi** and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that **onsemi** was negligent regarding the design or manufacture of the part. **onsemi** is an Equal Opportunity/Affirmative Action Employer. This literature is subject to all applicable copyright laws and is not for resale in any manner.

ADDITIONAL INFORMATION

TECHNICAL PUBLICATIONS:
Technical Library: www.onsemi.com/design/resources/technical−documentation
onsemi Website: www.onsemi.com

ONLINE SUPPORT: www.onsemi.com/support
For additional information, please contact your local Sales Representative at
www.onsemi.com/support/sales